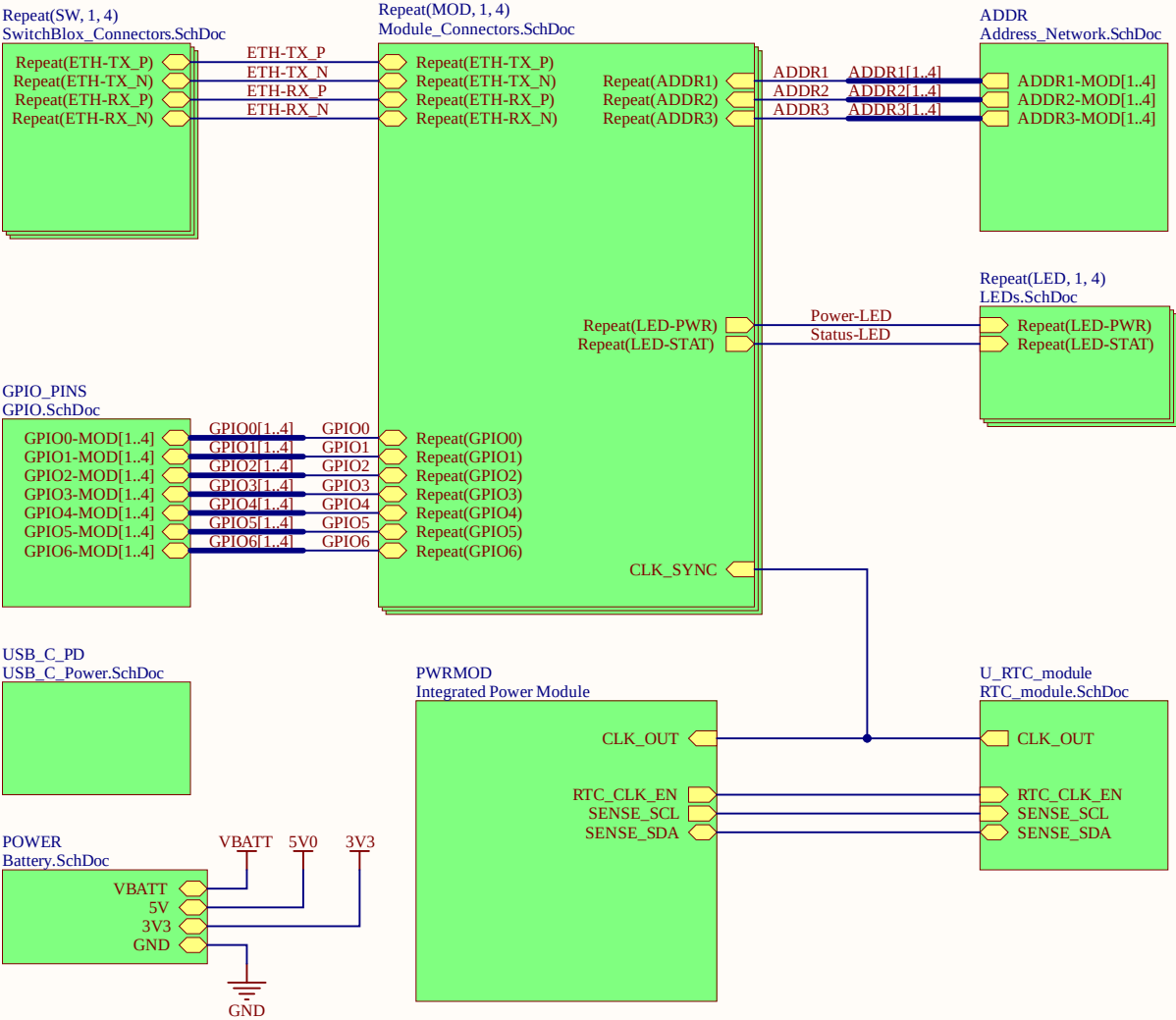
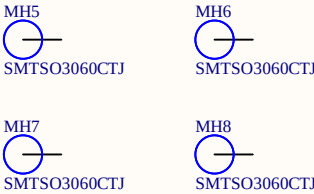


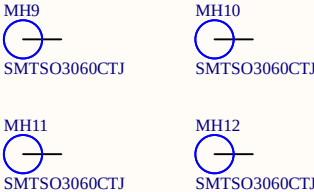
TODO:
VDD-3V3 Net changed in power mod, possible conflict with backplane hierarchy?

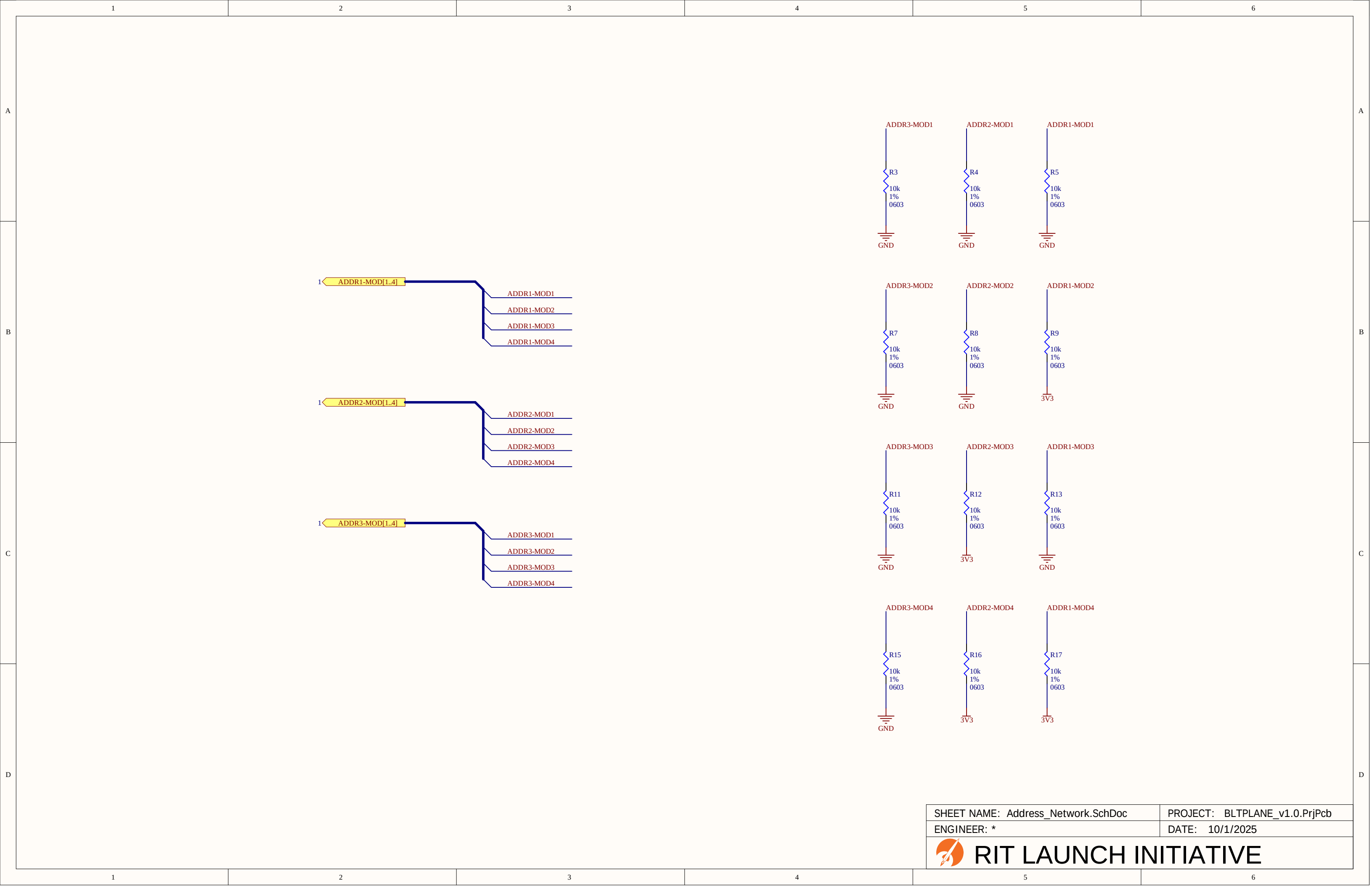


MOD1-3 Mounting Holes



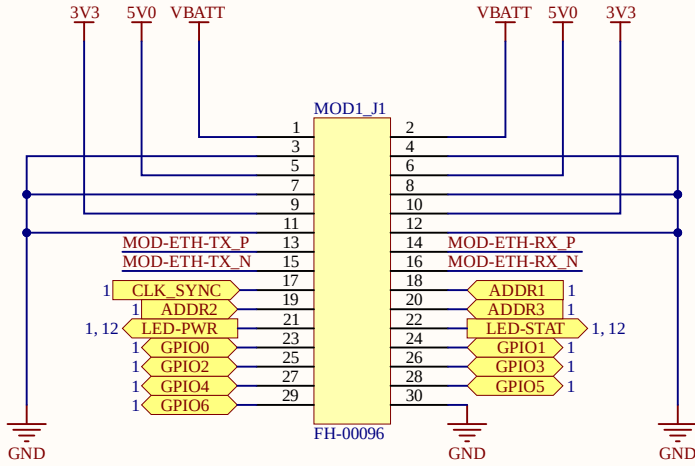
MOD2-4 Mounting Holes



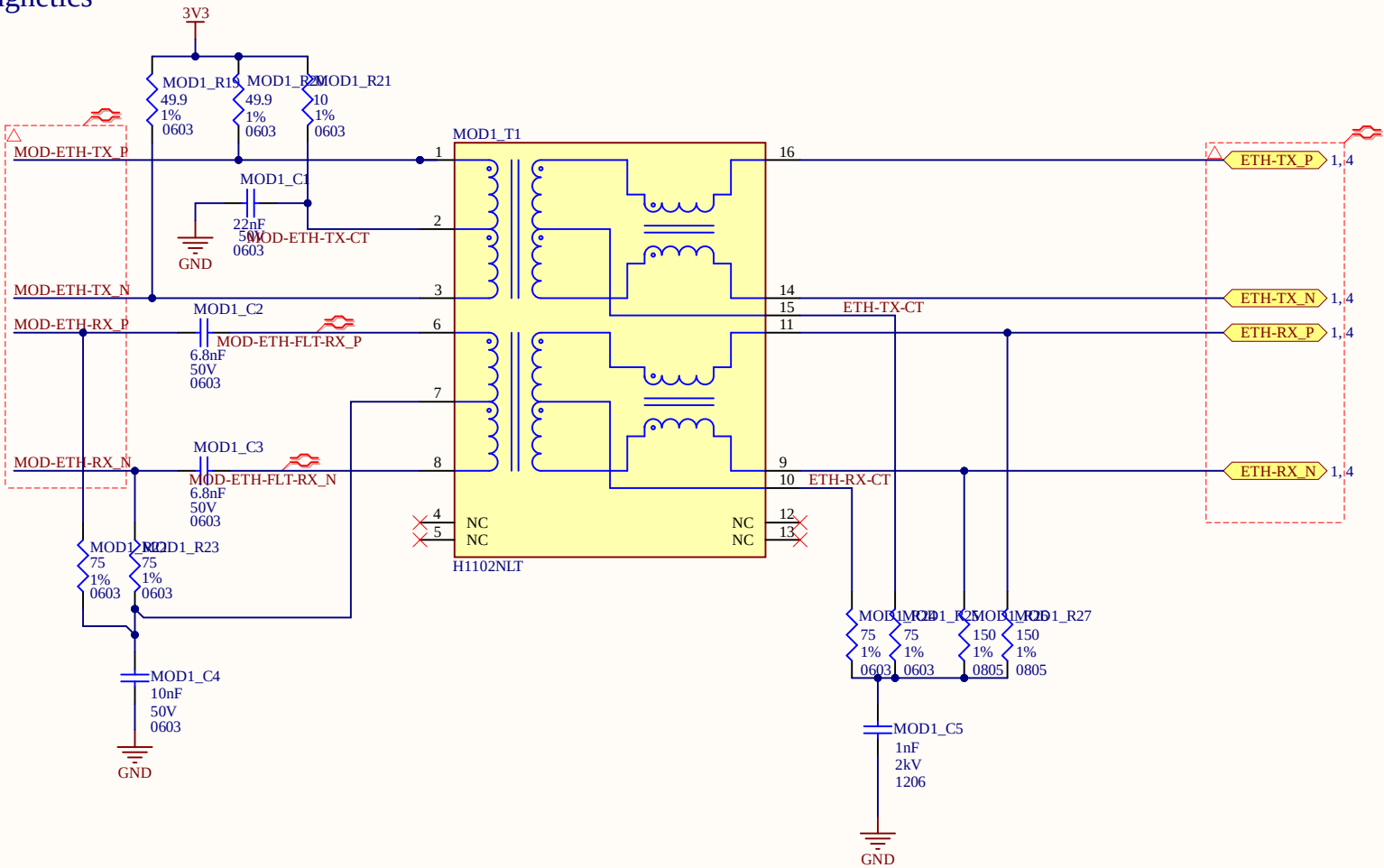


Module Connector

Description	PIN#		Description
Battery Voltage	1	2	Battery Voltage
GND	3	4	GND
5v	5	6	5v
GND	7	8	GND
3v3	9	10	3v3
GND	11	12	GND
TX+	13	14	RX+
TX-	15	16	RX-
Clock Sync	17	18	ADDR1
ADDR 2	19	20	ADDR 3
Power LED	21	22	Status LED
GPIO 0	23	24	GPIO 1
GPIO 2	25	26	GPIO 3
GPIO 4	27	28	GPIO 5
GPIO 6	29	30	GPIO 7

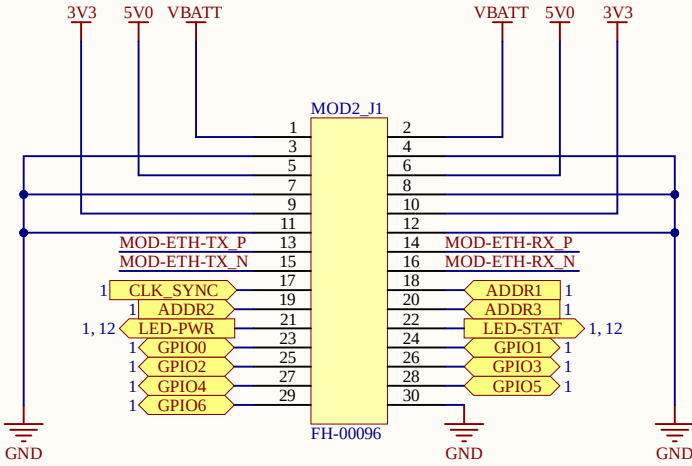


Ethernet Magnetics

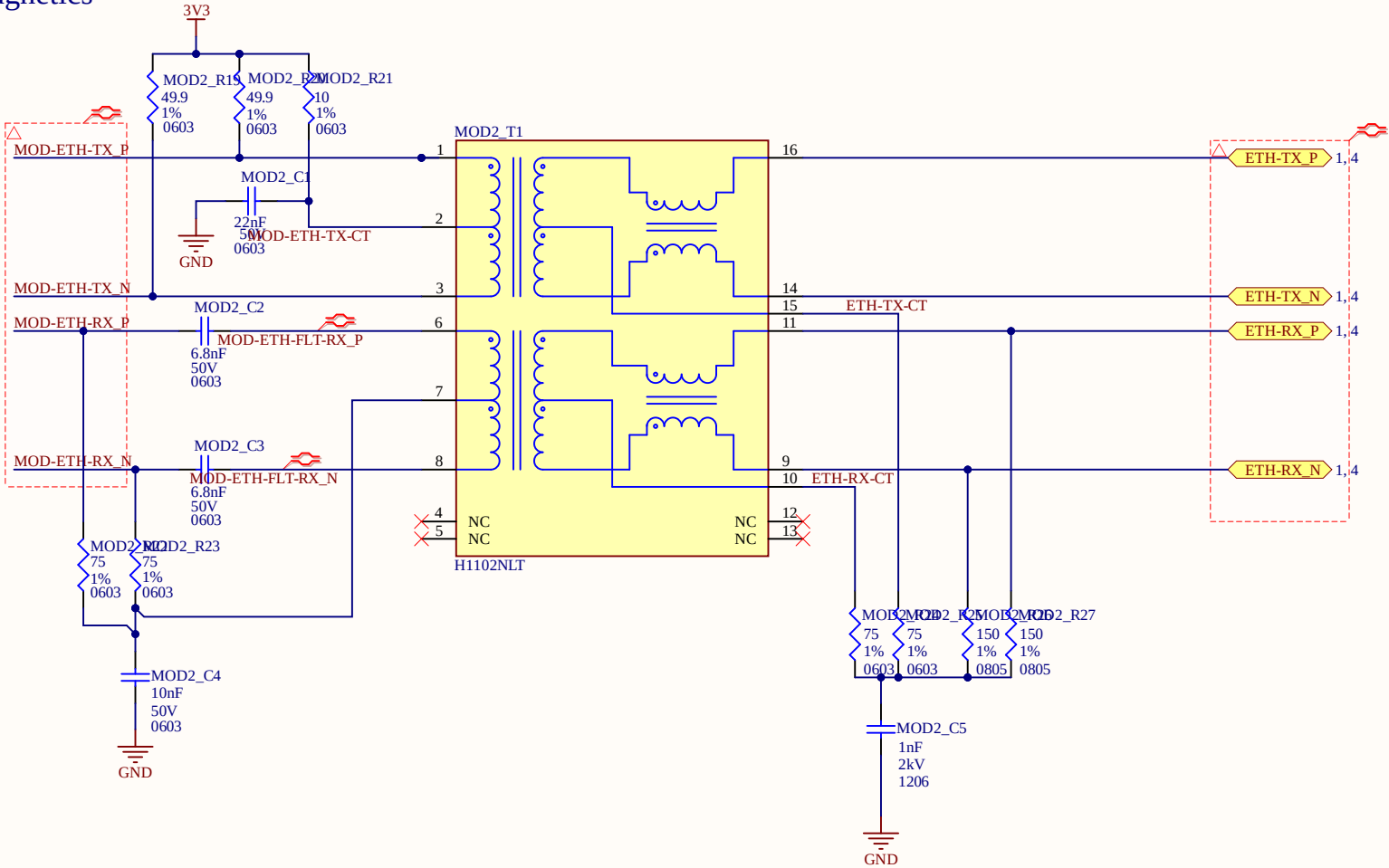


Module Connector

Description	PIN#	Description
Battery Voltage	1	2 Battery Voltage
GND	3	4 GND
5v	5	6 5v
GND	7	8 GND
3v3	9	10 3v3
GND	11	12 GND
TX+	13	14 RX+
TX-	15	16 RX-
Clock Sync	17	18 ADDR1
ADDR 2	19	20 ADDR 3
Power LED	21	22 Status LED
GPIO 0	23	24 GPIO 1
GPIO 2	25	26 GPIO 3
GPIO 4	27	28 GPIO 5
GPIO 6	29	30 GPIO 7

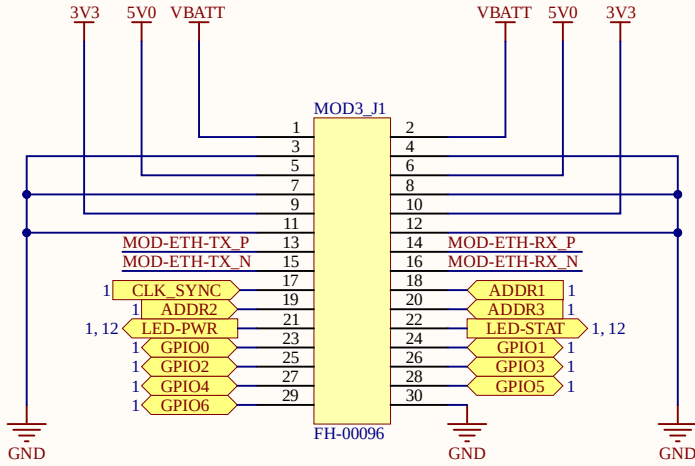


Ethernet Magnetics

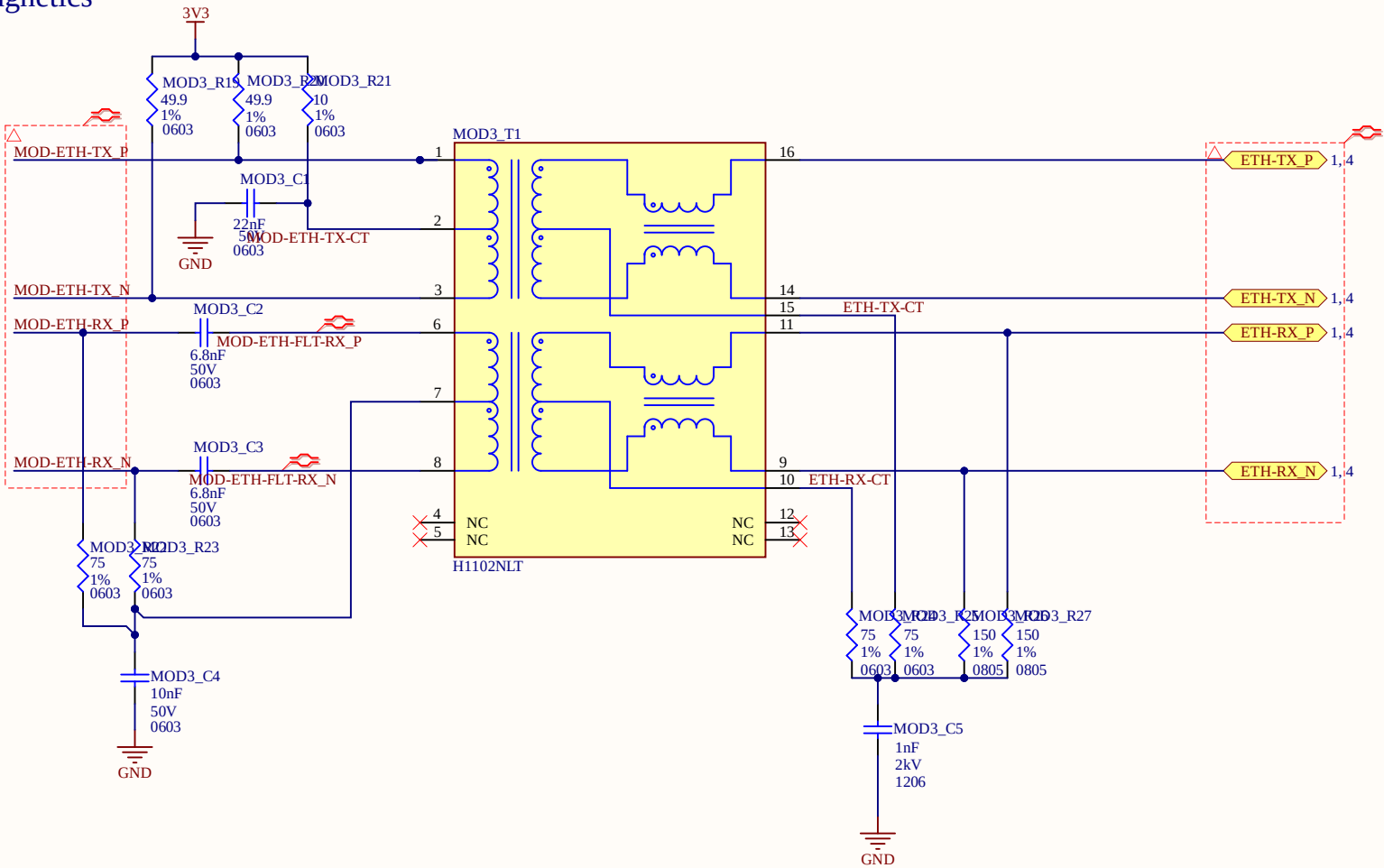


Module Connector

Description	PIN#		Description
Battery Voltage	1	2	Battery Voltage
GND	3	4	GND
5v	5	6	5v
GND	7	8	GND
3v3	9	10	3v3
GND	11	12	GND
TX+	13	14	RX+
TX-	15	16	RX-
Clock Sync	17	18	ADDR1
ADDR 2	19	20	ADDR 3
Power LED	21	22	Status LED
GPIO 0	23	24	GPIO 1
GPIO 2	25	26	GPIO 3
GPIO 4	27	28	GPIO 5
GPIO 6	29	30	GPIO 7

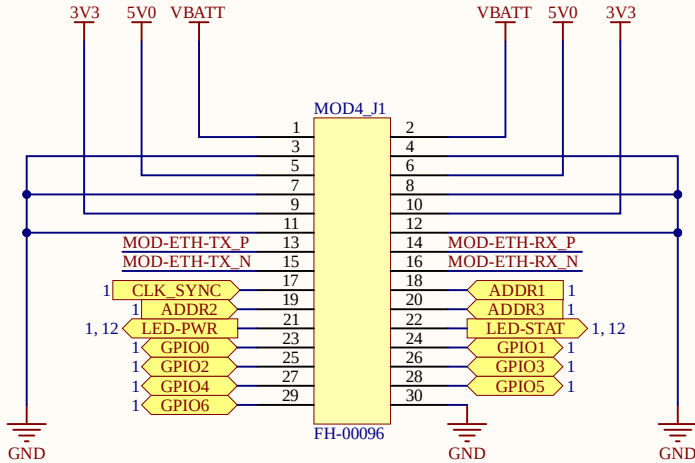


Ethernet Magnetics

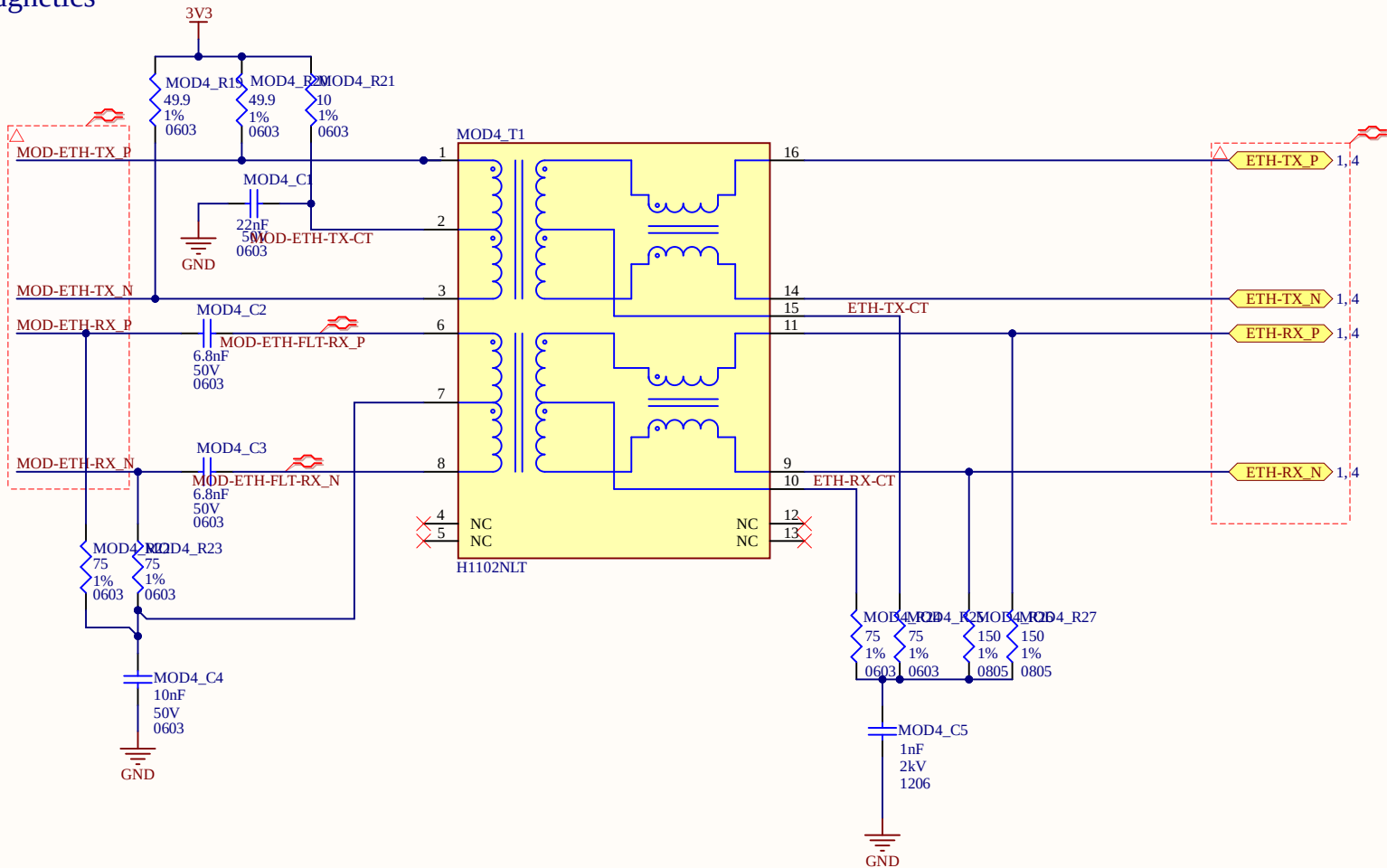


Module Connector

Description	PIN#		Description
Battery Voltage	1	2	Battery Voltage
GND	3	4	GND
5v	5	6	5v
GND	7	8	GND
3v3	9	10	3v3
GND	11	12	GND
TX+	13	14	RX+
TX-	15	16	RX-
Clock Sync	17	18	ADDR1
ADDR 2	19	20	ADDR 3
Power LED	21	22	Status LED
GPIO 0	23	24	GPIO 1
GPIO 2	25	26	GPIO 3
GPIO 4	27	28	GPIO 5
GPIO 6	29	30	GPIO 7



Ethernet Magnetics

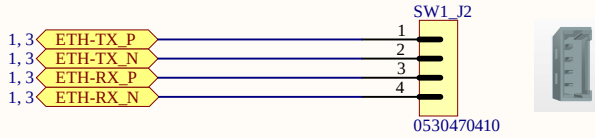
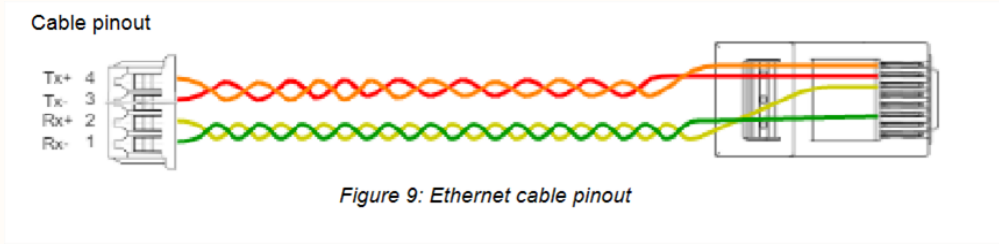


- spec cables

Ethernet net names are from the perspective of the modules.

The Switchblox supports Auto-MDIX crossover so it is okay that TX goes to TX and RX to RX

Pinout is flipped as the cables cross over signals

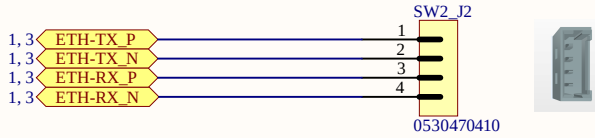
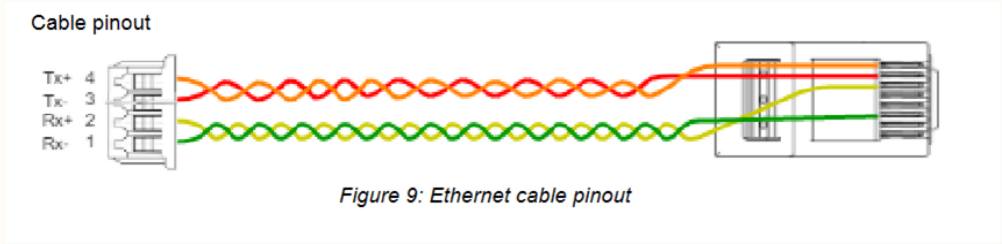


- spec cables

Ethernet net names are from the perspective of the modules.

The Switchblox supports Auto-MDIX crossover so it is okay that TX goes to TX and RX to RX

Pinout is flipped as the cables cross over signals

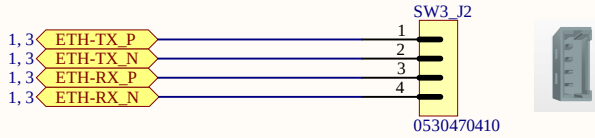
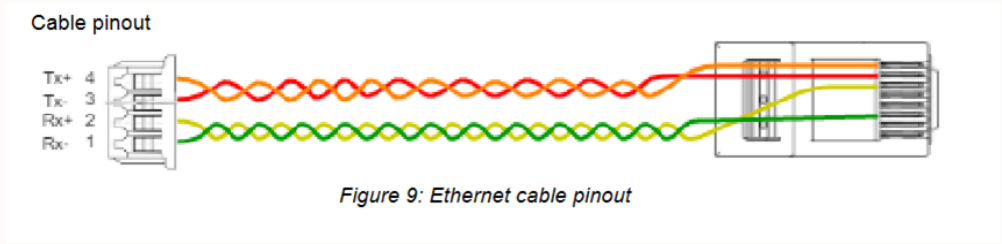


- spec cables

Ethernet net names are from the perspective of the modules.

The Switchblox supports Auto-MDIX crossover so it is okay that TX goes to TX and RX to RX

Pinout is flipped as the cables cross over signals

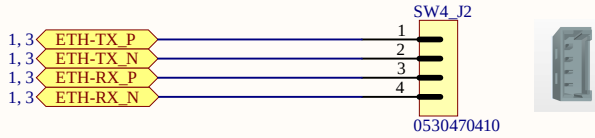
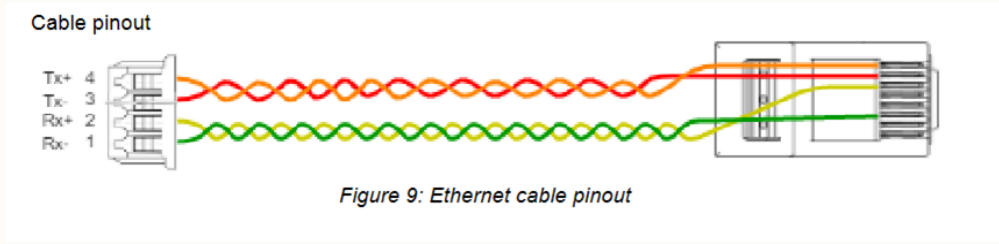


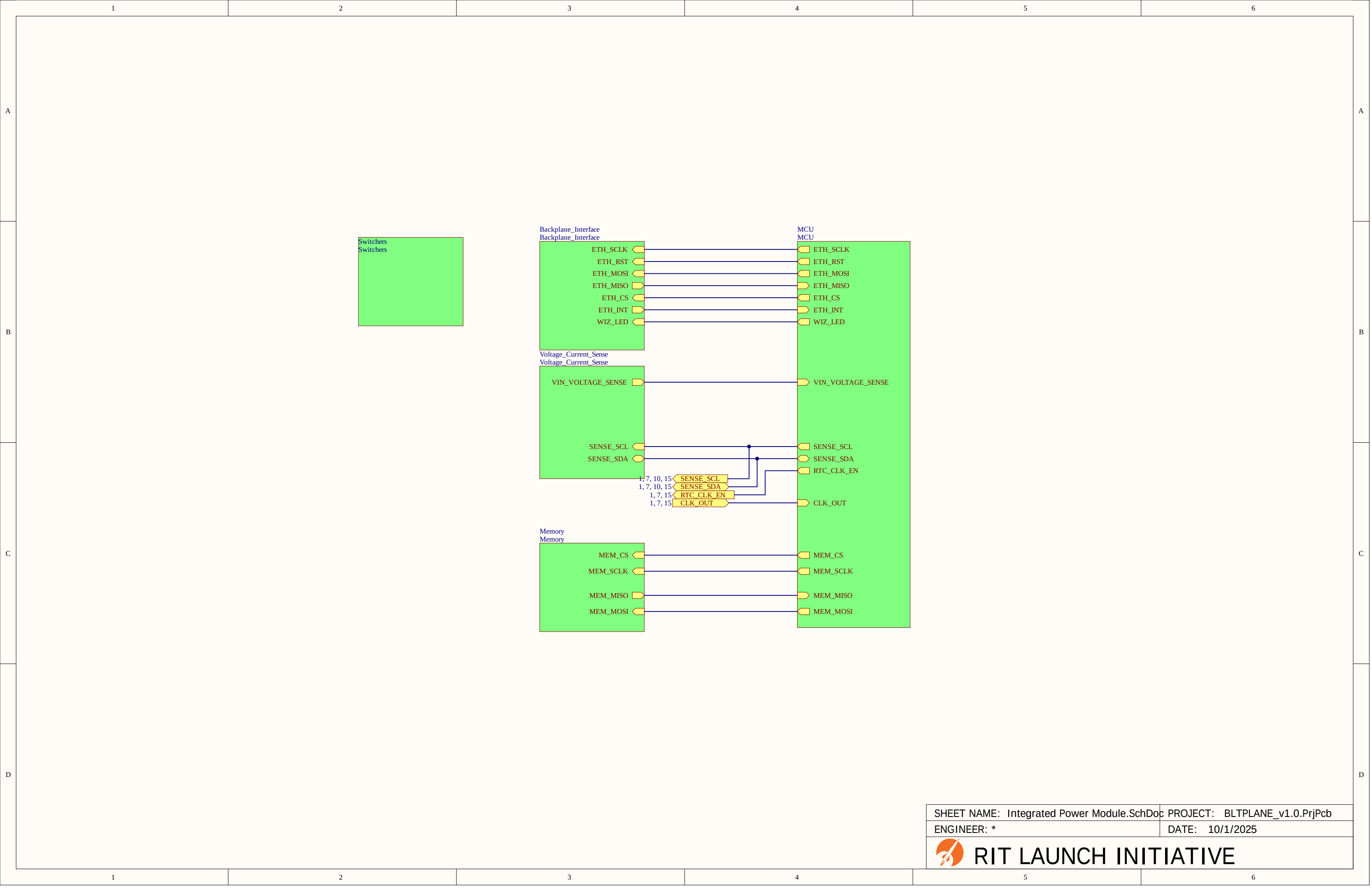
- spec cables

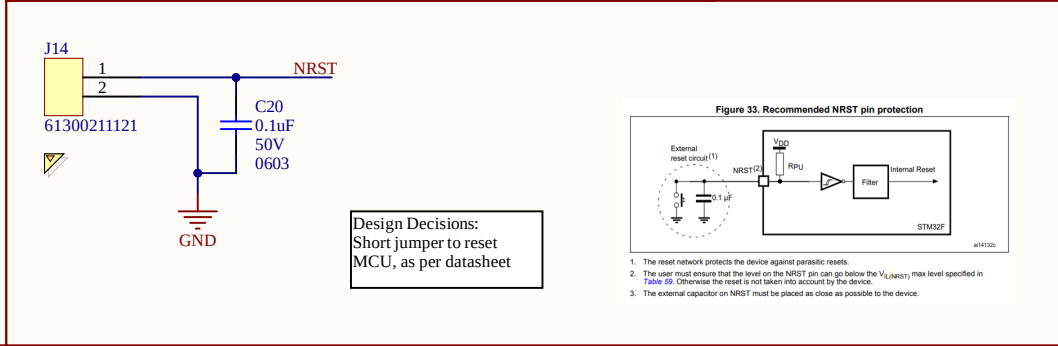
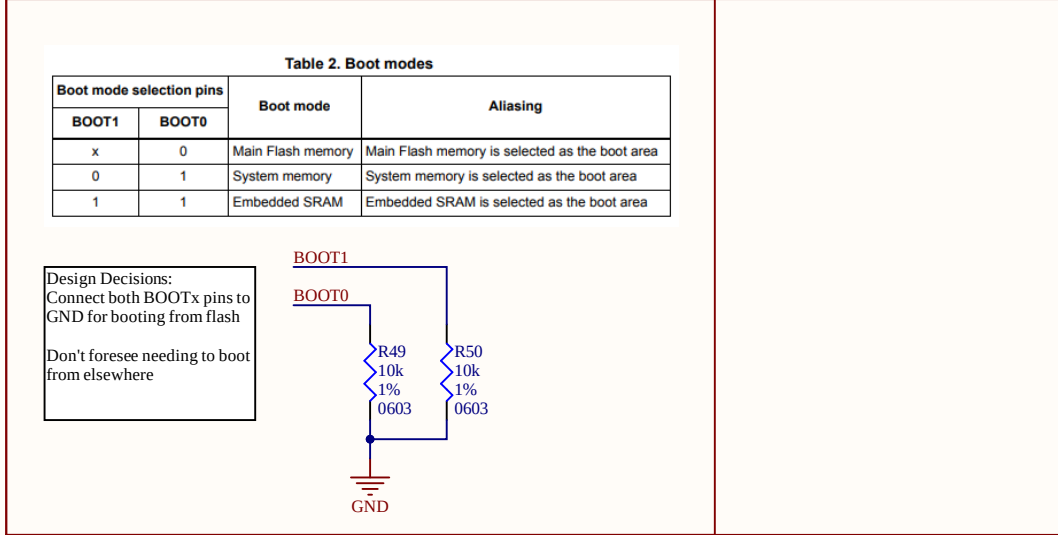
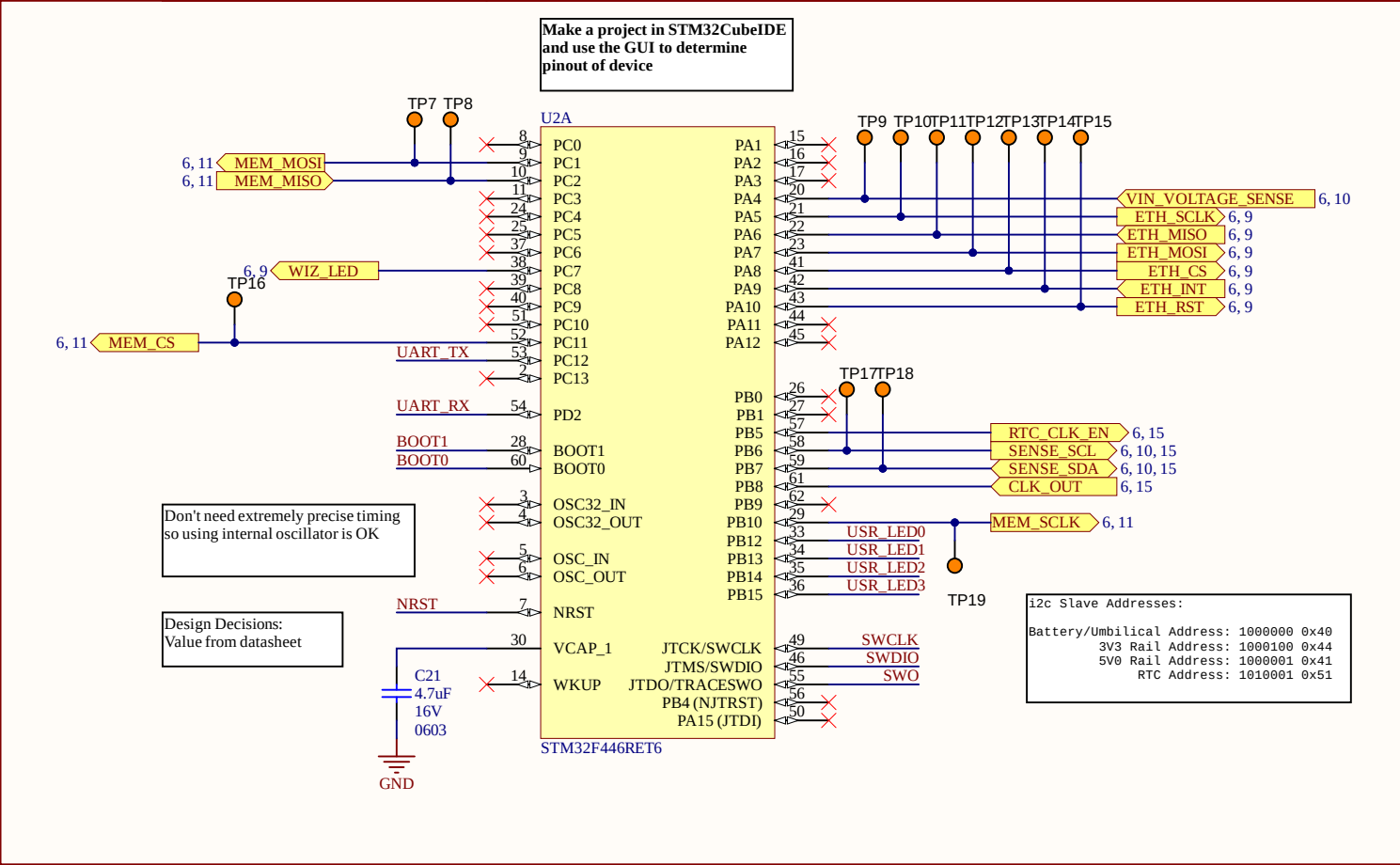
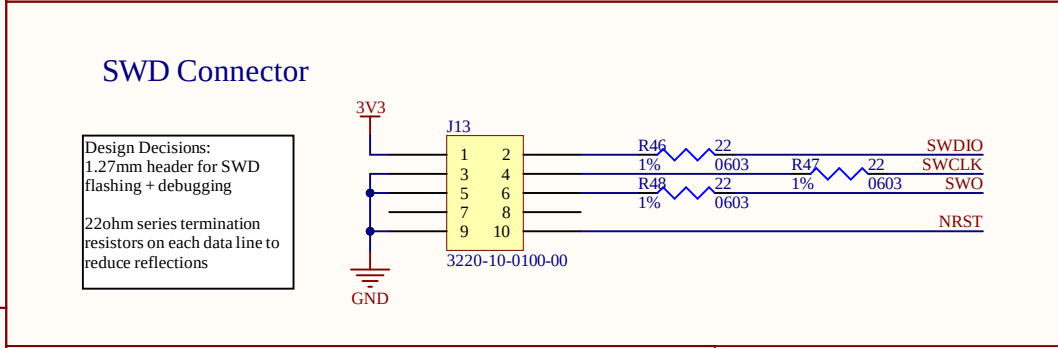
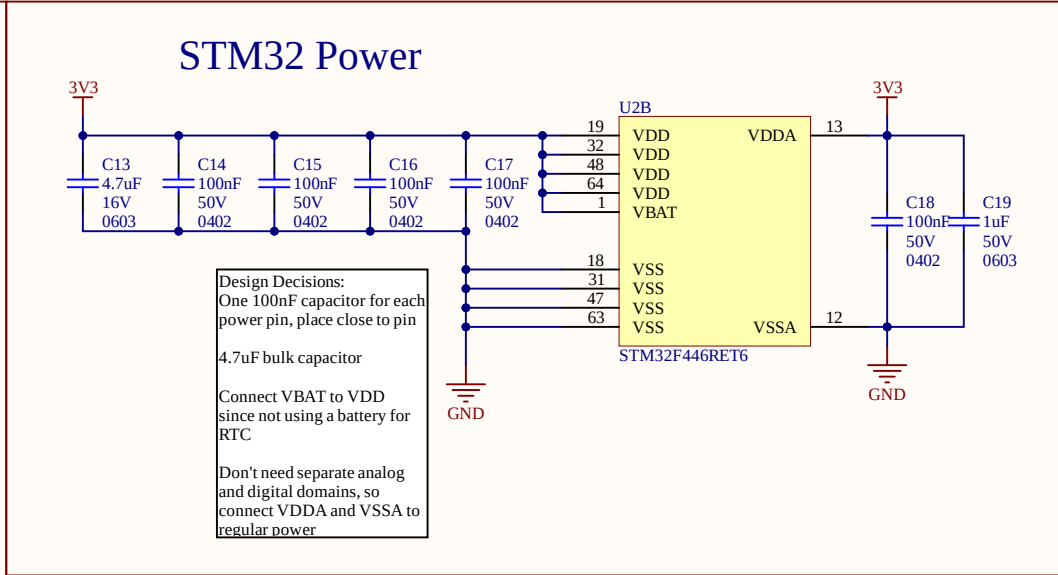
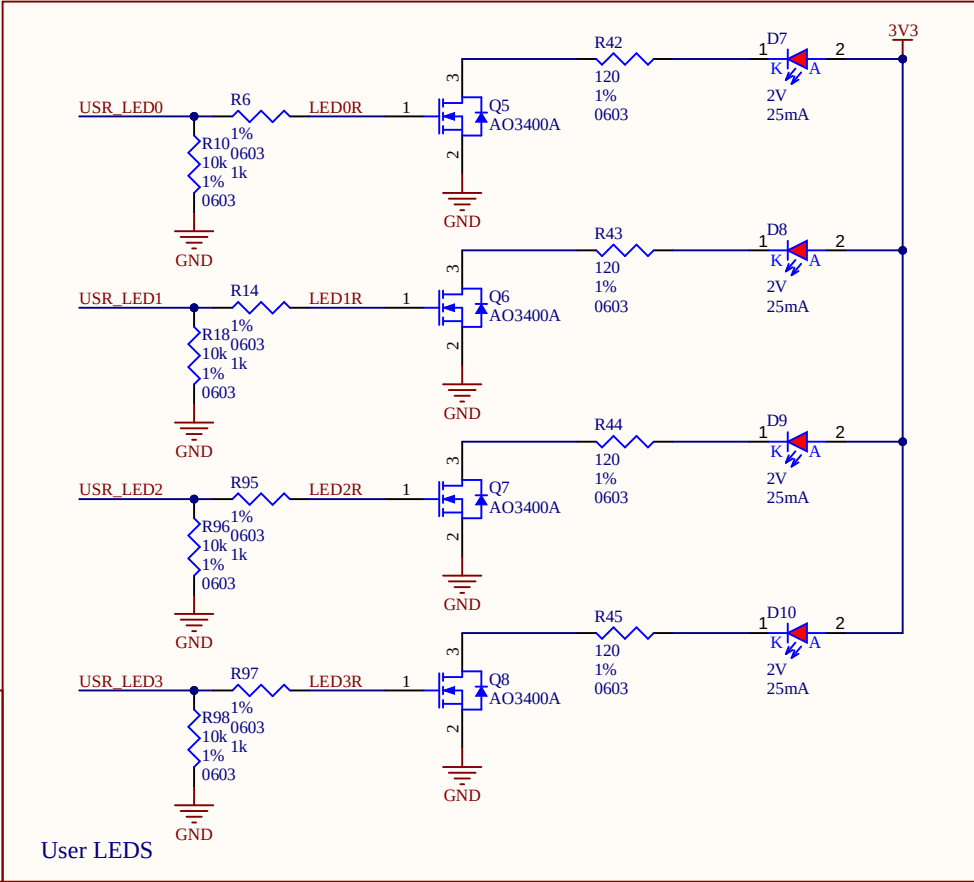
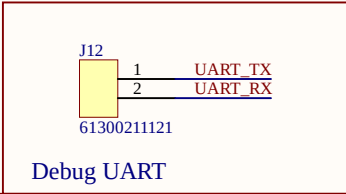
Ethernet net names are from the perspective of the modules.

The Switchblox supports Auto-MDIX crossover so it is okay that TX goes to TX and RX to RX

Pinout is flipped as the cables cross over signals



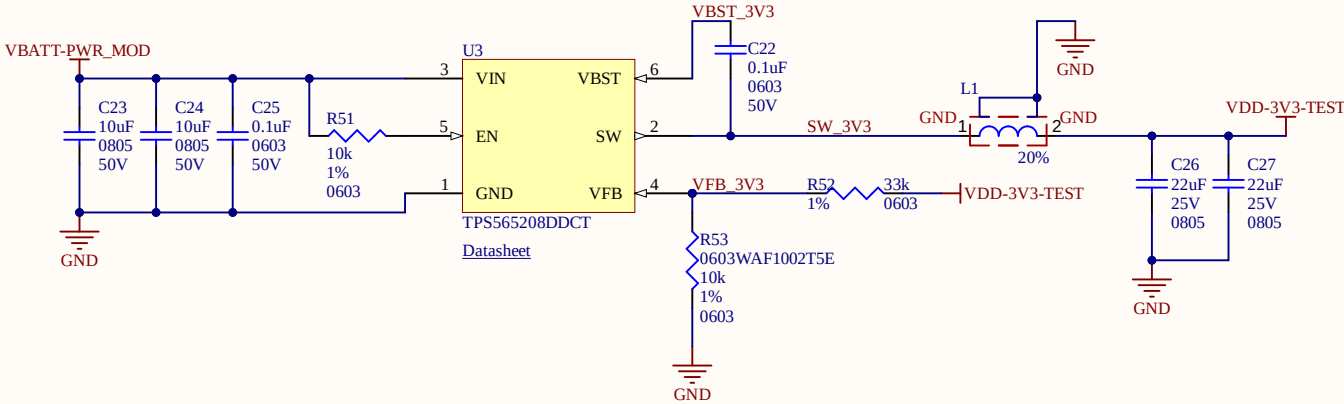




Protections:
-UVLO: shutdown if VIN < 3.6 (non-latching)
-OCL: shutdown if IOUT > 6.7A
-VOUT: shutdown if VOUT +-65% nominal
-Thermal: shutdown if T > 172C

Switching Frequency: 550 kHz

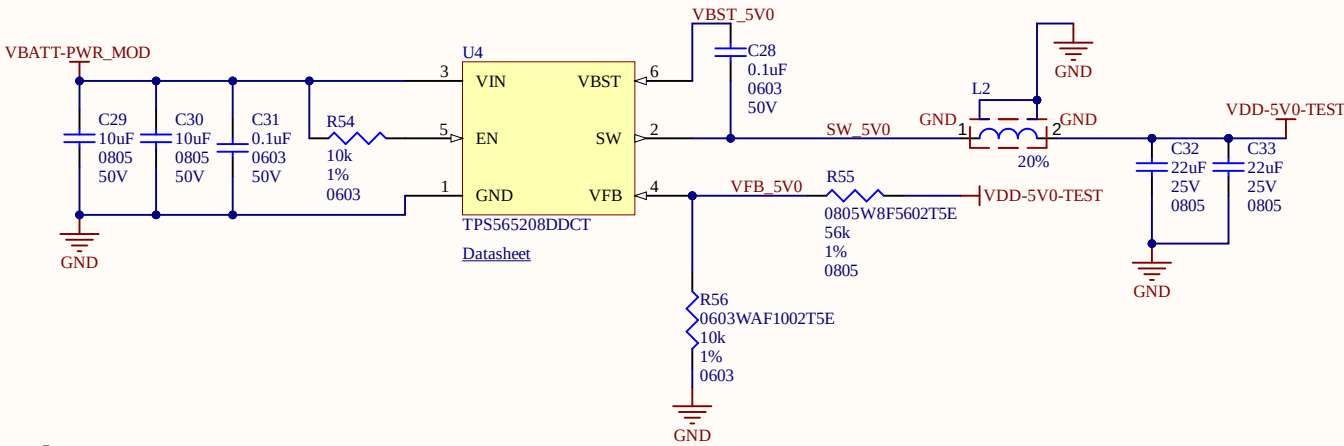
[Simulation](#)



3V3 Switcher

Switching Frequency: 550 kHz

[Simulation](#)



5V Switcher

- 5-A Maximum Output Current
- Integrated 31-mΩ and 16-mΩ FETs
- D-CAP2™ Mode Control with Fast Transient Response
- Input Voltage Range: 4.5 V to 17 V
- Output Voltage Range: 0.76 V to 7 V
- Continuous Current Mode
- 500-kHz Switching Frequency
- Low Shutdown Current of Less than 1 μA
- 1% Feedback Voltage Accuracy
- Startup from Pre-biased Output Voltage
- Cycle-by-Cycle Current Limit
- Hiccup-mode Overcurrent Protection
- Non-Latch UVP and TSD Protections
- Fixed Soft Start: 1.0 ms

8.2 Typical Application

The application schematic in [Figure 14](#) shows the TPS565208 4.5-V to 17-V input, 1.05-V output converter design meeting the requirements for 5-A output. This circuit is available as the evaluation module (EVM). The sections provide the design procedure.

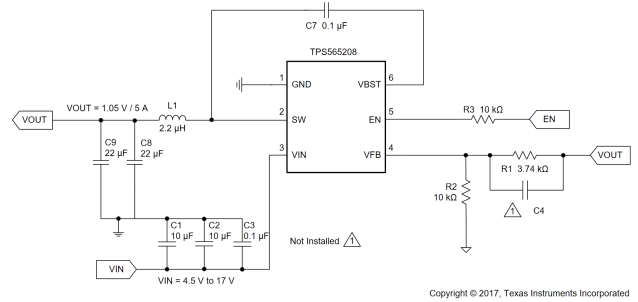


Table 2. Recommended Component Values

OUTPUT VOLTAGE (V)	R1 (kΩ)	R2 (kΩ)	L1 (μH)			C8 + C9 (μF)
			MIN	TYP	MAX	
1	3.09	10.0	1	2.2	4.7	20 to 68
1.05	3.74	10.0	1	2.2	4.7	20 to 68
1.2	5.76	10.0	1	2.2	4.7	20 to 68
1.5	9.53	10.0	1.5	2.2	4.7	20 to 68
1.8	13.7	10.0	1.5	2.2	4.7	20 to 68
2.5	22.6	10.0	2.2	2.2	4.7	20 to 68
3.3	33.2	10.0	2.2	2.2	4.7	20 to 68
5	54.9	10.0	3.3	3.3	4.7	20 to 68
6.5	75	10.0	3.3	3.3	4.7	20 to 68

SHEET NAME: Switchers.SchDoc

PROJECT: BLTPLANE_v1.0.PrjPcb

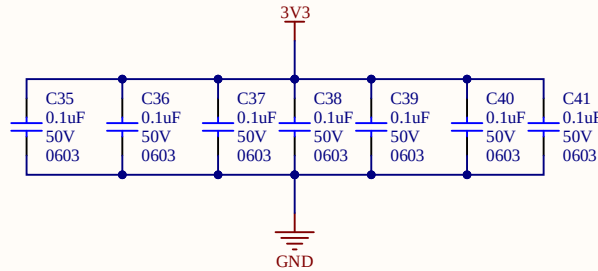
ENGINEER: *

DATE: 10/1/2025



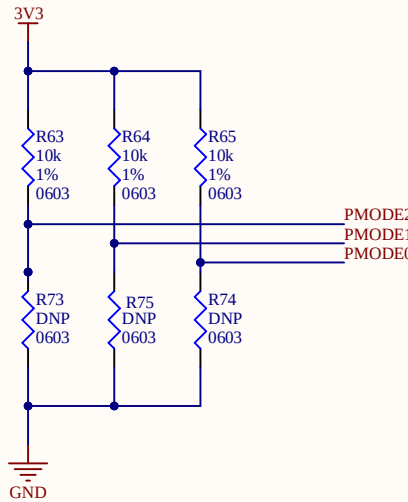
RIT LAUNCH INITIATIVE

Decoupling Caps

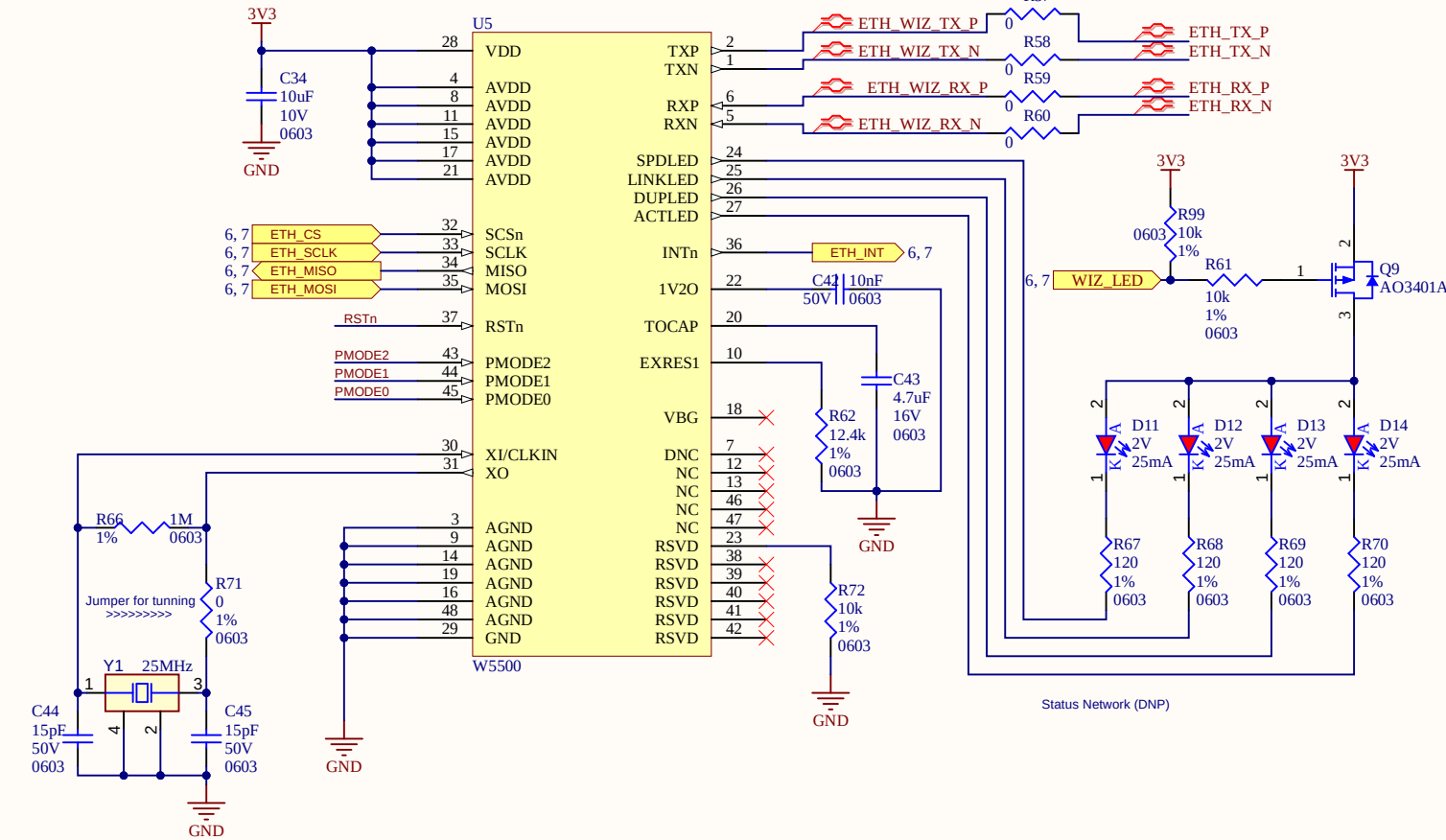


PMODE Select

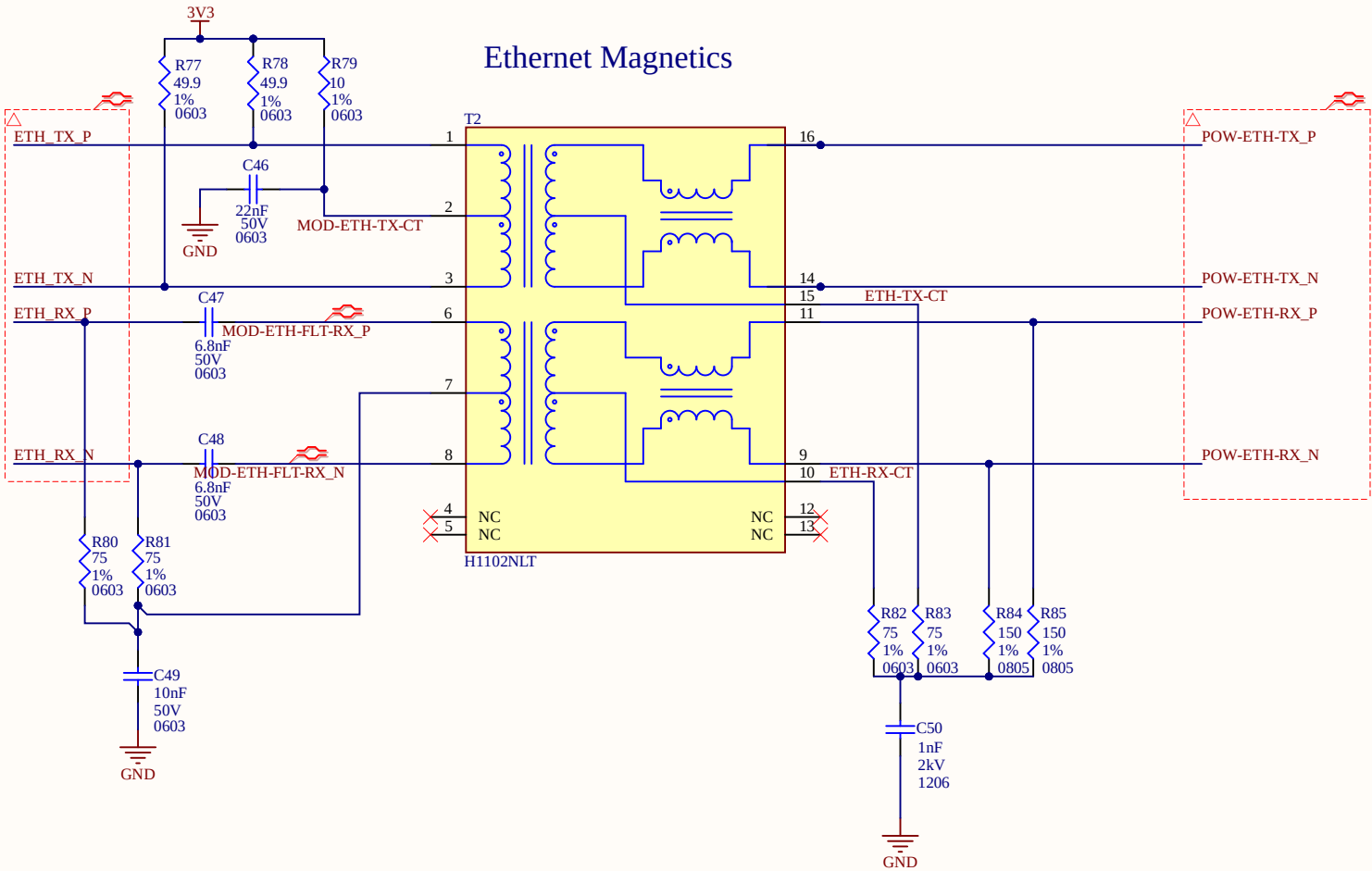
PMODE [2:0]			Description
2	1	0	
0	0	0	10BT Half-duplex, Auto-negotiation disabled
0	0	1	10BT Full-duplex, Auto-negotiation disabled
0	1	0	100BT Half-duplex, Auto-negotiation disabled
0	1	1	100BT Full-duplex, Auto-negotiation disabled
1	0	0	100BT Half-duplex, Auto-negotiation enabled
1	0	1	Not used
1	1	0	Not used
1	1	1	All capable, Auto-negotiation enabled



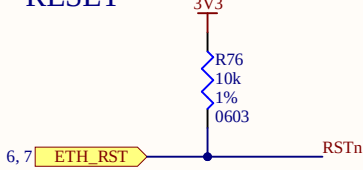
WizNet 5500



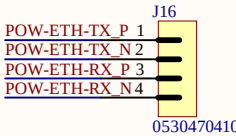
Ethernet Magnetics



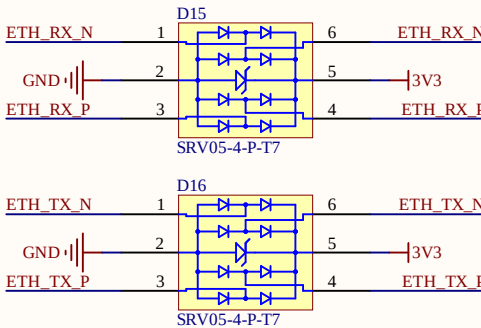
RESET



JST Connector for Power Mod



ESD Protection



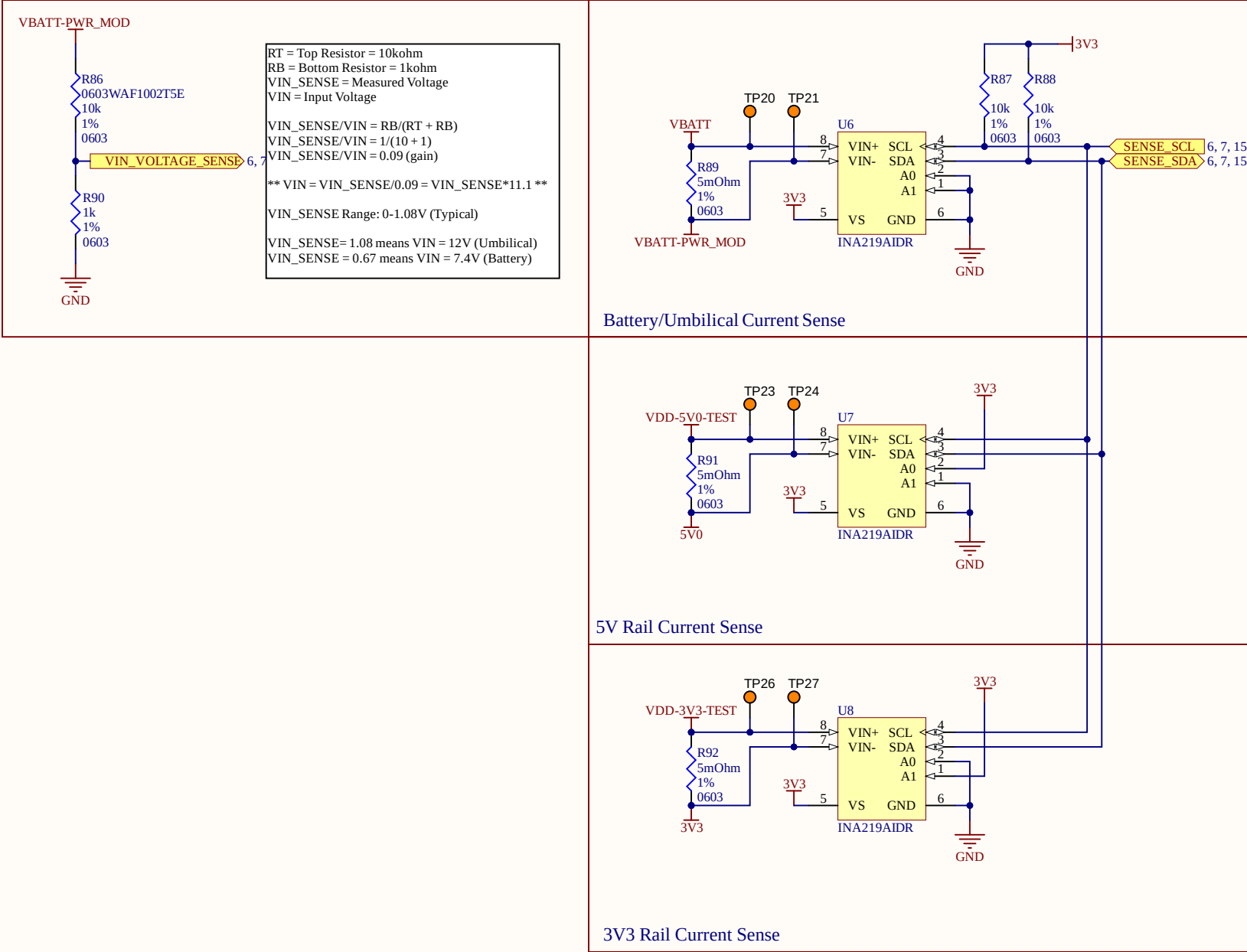


Table 1. INA219 Address Pins and Slave Addresses		
A1	A0	SLAVE ADDRESS
GND	GND	1000000
GND	VS+	1000001
GND	SDA	1000010
GND	SCL	1000011
VS+	GND	1000100
VS+	VS+	1000101
VS+	SDA	1000110
VS+	SCL	1000111
SDA	GND	1001000
SDA	VS+	1001001
SDA	SDA	1001010
SDA	SCL	1001011
SCL	GND	1001100
SCL	VS+	1001101
SCL	SDA	1001110
SCL	SCL	1001111

i2c Slave Addresses:

Battery/Umbilical Address: 1000000 0x40
3V3 Rail Address: 1000100 0x44
5V0 Rail Address: 1000001 0x41
RTC Address: 1010001 0x51

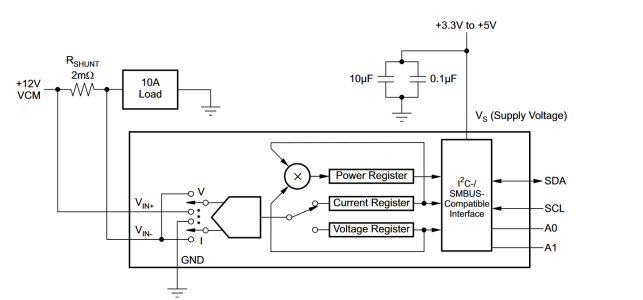
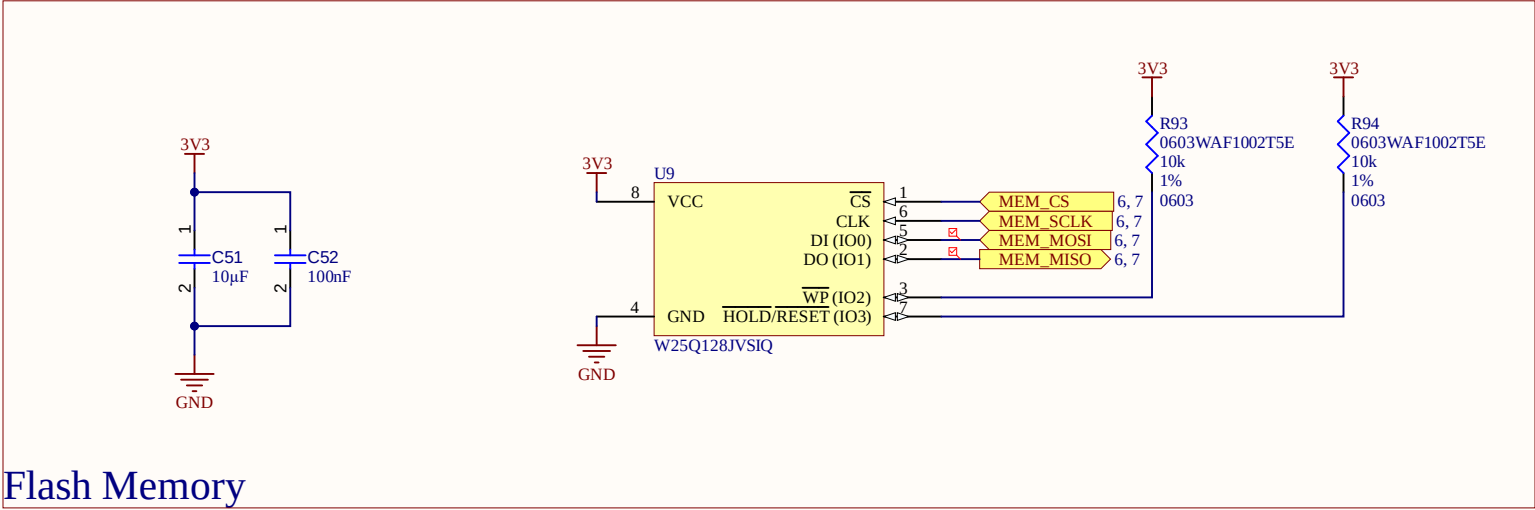
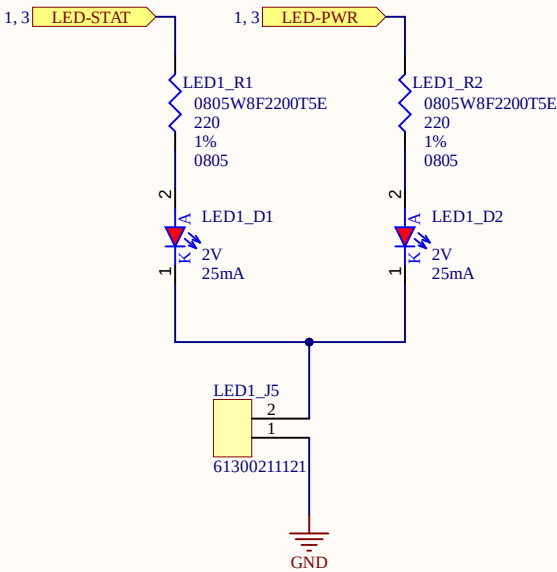
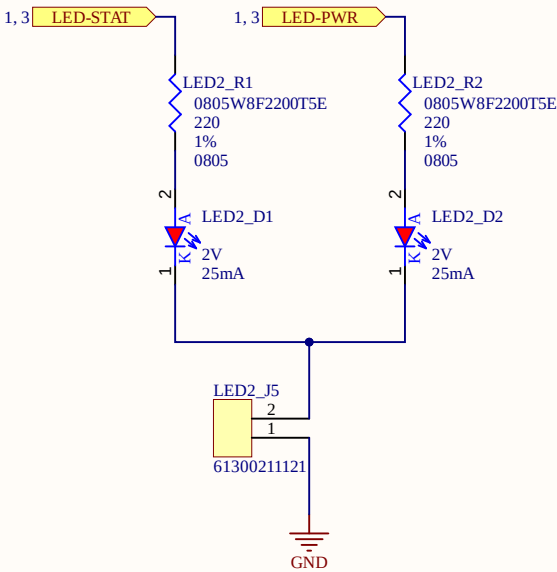


Figure 29. Example Circuit Configuration

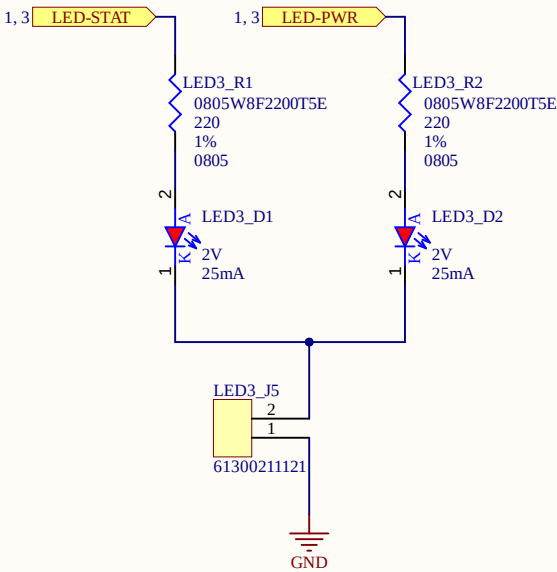




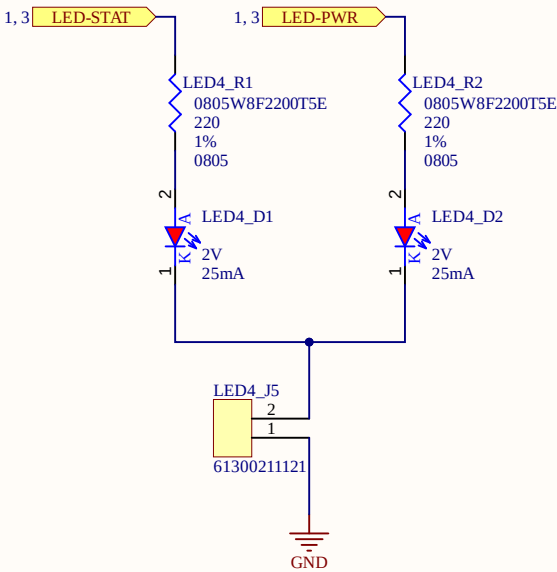
$V_f = 2V$
 $V_r = 1.3V$
 $I_f = I_r = 6mA$
 $R = V_r / I_r = 1.3 / 0.006 = 216 \approx 220$



$V_f = 2V$
 $V_r = 1.3V$
 $I_f = I_r = 6mA$
 $R = V_r / I_r = 1.3 / 0.006 = 216 = 220$



$V_f = 2V$
 $V_r = 1.3V$
 $I_f = I_r = 6mA$
 $R = V_r / I_r = 1.3 / 0.006 = 216 = 220$



$V_f = 2V$
 $V_r = 1.3V$
 $I_f = I_r = 6mA$
 $R = V_r / I_r = 1.3 / 0.006 = 216 = 220$

Figure 3. EZ-PD BCR based Application Diagram (for Electronic Systems Requiring 12 V to 15 V Input at 2 A)

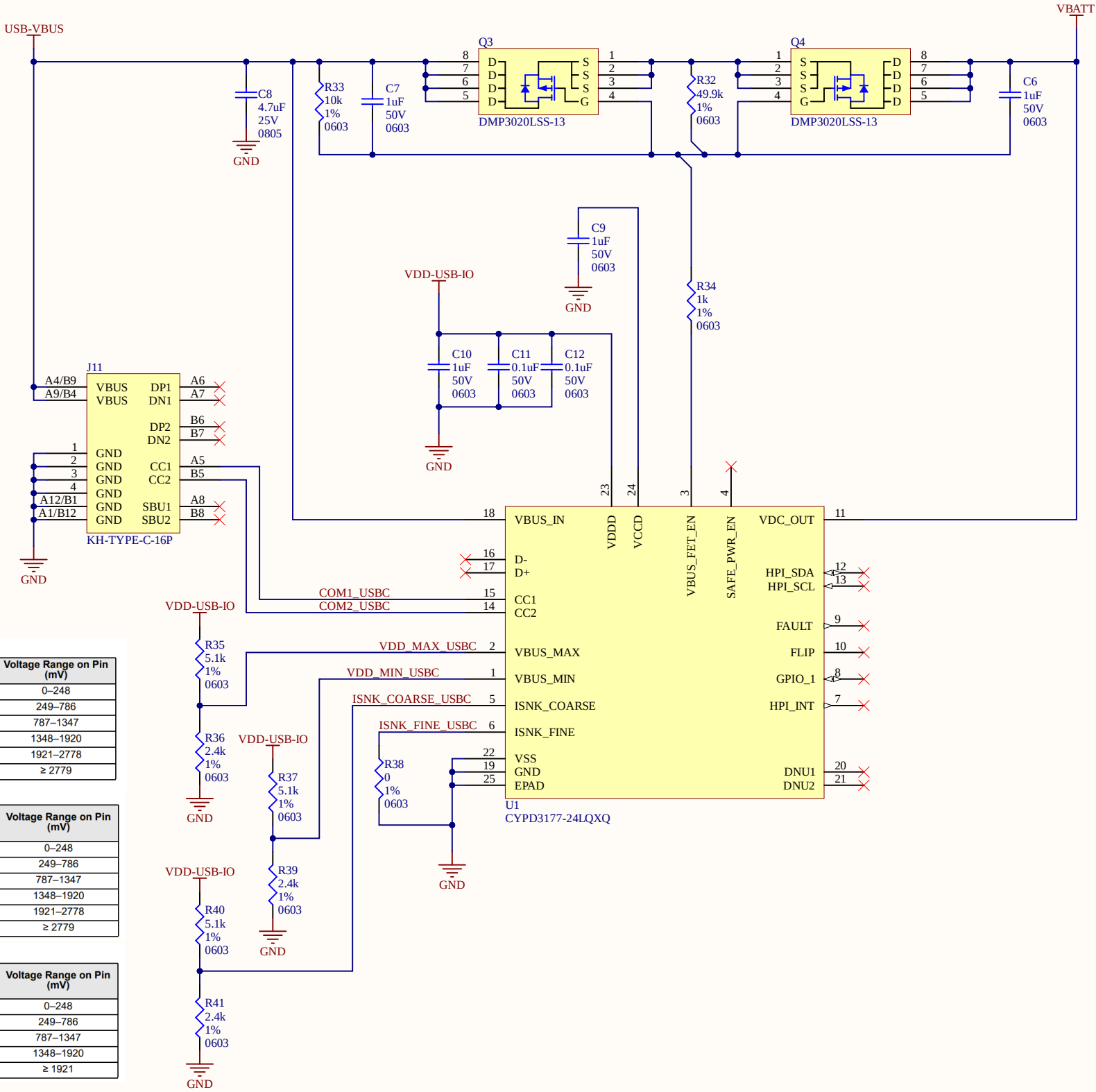
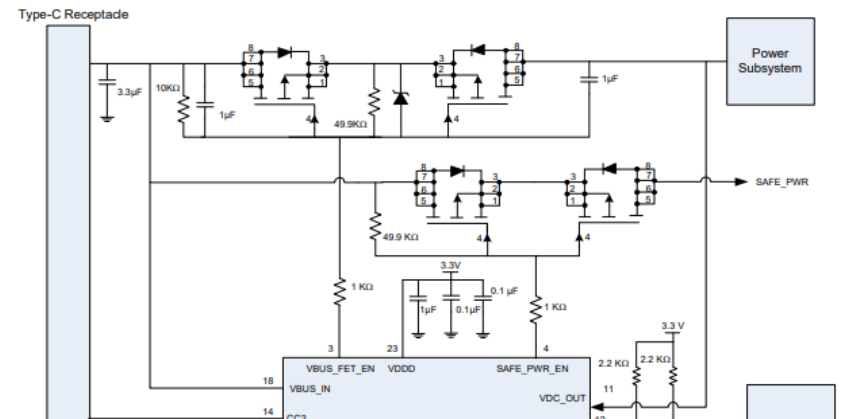


Table 2. Resistor Divider Values for Minimum or Maximum Voltage Requested on VBUS				
Voltage Requested (V)	Resistor Ratio Relative to VDDD = 3.3 V	Suggested Pull-up Resistor value (kΩ)	Suggested Pull-down Resistor value (kΩ)	Voltage Range on Pin (mV)
5	0/6	Open	0	0–248
9	1/6	5.1	1	249–786
12	2/6	5.1	2.4	787–1347
15	3/6	5.1	5.1	1348–1920
19	4/6	5.1	10	1921–2778
20	≥ 5/6	0	Open	≥ 2779

Table 3. Resistor Divider Values for Coarse Setting on Operating Current (For VDDD = 3.3 V)				
Operating Current Requested for Coarse Setting (A)	Resistor Ratio Relative to VDDD = 3.3 V	Suggested Pull-up Resistor Value (kΩ)	Suggested Pull-down Resistor Value (kΩ)	Voltage Range on Pin (mV)
0	0/6	Open	0	0–248
1	1/6	5.1	1	249–786
2	2/6	5.1	2.4	787–1347
3	3/6	5.1	5.1	1348–1920
4	4/6	5.1	10	1921–2778
5	≥ 5/6	0	Open	≥ 2779

Table 4. Resistor Divider Values for Fine Setting on Operating Current (For VDDD = 3.3 V)				
Operating Current Requested for Fine Setting (A)	Resistor Ratio Relative to VDDD = 3.3 V	Suggested Pull-up Resistor Value (kΩ)	Suggested Pull-down Resistor Value (kΩ)	Voltage Range on Pin (mV)
+0	0/6	Open	0	0–248
+250	1/6	5.1	1	249–786
+500	2/6	5.1	2.4	787–1347
+750	3/6	5.1	5.1	1348–1920
+900	≥ 4/6	0	Open	≥ 1921

